

- 1 (a) change in velocity of the body is always perpendicular to velocity when speed is constant. B1
- acceleration and so resultant force [Newton's second] is always perpendicular to the velocity B1
- velocity is tangent to circular path, so resultant force (perpendicular to the velocity) directed towards centre of circle B1
- (b) (i) centripetal acceleration
- $$= \frac{\left(\frac{25 \times 10^3}{60 \times 60}\right)^2}{7.0} \quad \text{M1}$$
- $= 6.9 \text{ m s}^{-2} \text{ or } 6.89 \text{ m s}^{-2} \quad \text{A1}$
- (ii) force on mass $= 0.50 \times 6.889 = 3.4445 \text{ N} \quad \text{C1}$
- displacement $= 3.4445 \times 5.0 = 17 \text{ mm or } 17.2 \text{ mm} \quad \text{A1}$
- (iii) extension of spring at B > spring at A / spring at B is extended while spring at A is compressed to provide this resultant force (towards B or centre) M1
- pointer moves towards A. A1

2 (a) The gravitational potential at a point is defined as the work done per unit mass in bringing a